

Lecture 23

Eigenvalue II: Spectral Decomposition

谱分解

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Today's Lecture: Outline

Main topic: Spectral Theorem

1. Eigenbasis
2. Spectral Theorem

After the lecture, you should be able to

1. Provide an example where eigenvectors do not form a basis
2. Describe the spectral decomposition and related properties
3. Explain the main proof steps of spectral decomposition

Review

5 mins

Eigenvalues and Eigenvectors

Definition 21.1 (Eigenvalues and Eigenvectors)

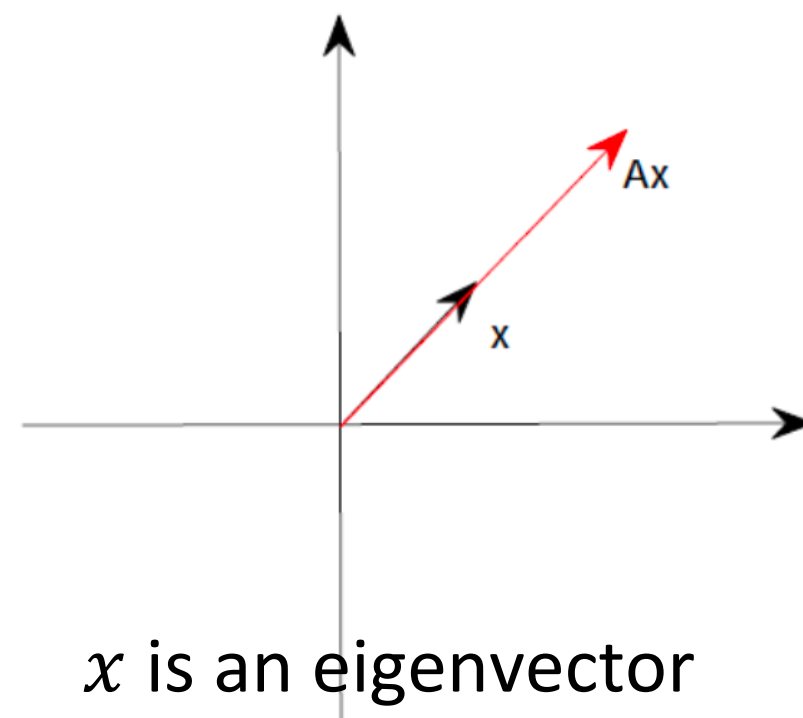
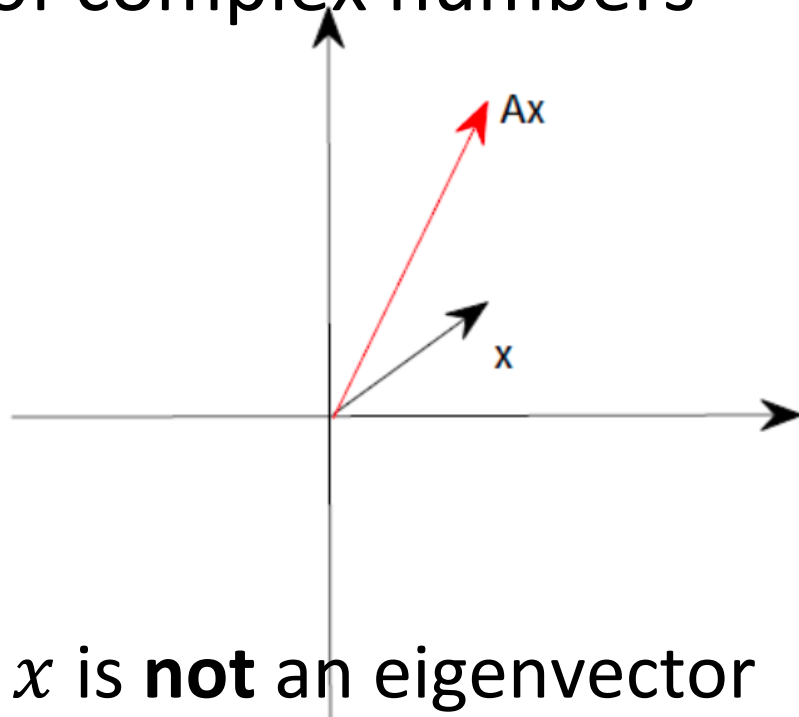
Let $A \in \mathbb{C}^{n \times n}$ be a square matrix.

If there exists a scalar λ ($\in \mathbb{R}$ or $\mathbb{C}$) and a **nonzero** vector x such that

$$Ax = \lambda x,$$

then λ is called a (real or complex) **eigenvalue** and x is called an **eigenvector** corresponding to (or associated with) λ .

$\mathbb{C} :=$ set of complex numbers



General Procedure to Find Eigenvalues/Eigenvectors

Step 1: Solve $\det(A - \lambda I) = 0$ and get n roots $\lambda_1, \dots, \lambda_n$

Solving single-var polynomial equation

Step 2: For each λ_i , find the eigenspace $N(\lambda_i I - A)$

Solving a number of (up to n) linear systems of equations.

Any nonzero vector in $N(\lambda_i I - A)$ is an eigenvector corresponding to λ_i .

$$N(\lambda_i I - A) = \left\{ \text{eigenvectors of } A \text{ with corresponding to the eigenvalue } \lambda_i \right\} \cup \{0\}$$

Eigenvalues and Eigenvectors

Characteristic polynomial $p_A(\lambda) = \det(\lambda I - A)$ is a degree- n polynomial.

Remark: The coefficient of λ^n is **1**.

Corollary of Fundamental Theorem of Algebra:

Any $n \times n$ (no matter real or complex) matrix A has exactly n **complex eigenvalues** (counting multiplicity).

Here, multiplicity of λ_j is the power k of the term $(\lambda - \lambda_j)$ in the decomposition of the characteristic polynomial $p_A(\lambda) = \prod_{j=1}^n (\lambda - \lambda_j)^{k_j}$.

Part 1 Eigenbasis

20 mins

Multiset [多重集]

Question: What did you buy today?

✓ **Answer 1 (Regular Set):**

I bought apples and oranges.

→ Represented as the set: {apple,orange}

A set only tracks *which items* you purchased, ignoring how many.

✓ **Answer 2 (Multiset):**

I bought 3 apples and 2 oranges.

→ Represented as the multiset:

#{apple,apple,apple,orange,orange}

A multiset tracks *both which items and how many of each* you purchased, by allowing repeated elements.

Multiset [多重集]

Multiset A multiset is a generalization of a set that allows multiple instances of its elements. Unlike a set, where each element appears at most once, a multiset permits repeated elements.

We use $\# \{a_1, a_2, \dots\}$ to denote a multiset.

Multiplicity: If an element appears k times in the multiset, then the **multiplicity** of the element in the multiset is k .

Multiset [多重集]

Multiset A multiset is a generalization of a set that allows multiple instances of its elements. Unlike a set, where each element appears at most once, a multiset permits repeated elements.

We use $\# \{a_1, a_2, \dots\}$ to denote a multiset.

Multiplicity: If an element appears k times in the multiset, then the **multiplicity** of the element in the multiset is k .

Examples:

In the multiset $\# \{a, a, b\}$, the element a has multiplicity 2, and b has multiplicity 1.

In the multiset $\# \{a, a, a, b, b, b\}$, a and b both have multiplicity 3.

The set $\{a, b\}$ contains only elements a and b .

Order does not matter: $\# \{a, a, b\}$ and $\# \{a, b, a\}$ denote the same multiset.

Implication of Multiset

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \text{ has eigenvalues } 1, 1, 2$$

$$B = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \text{ has eigenvalues } 1, 2$$

Can we say that **A** and **B** have the same eigenvalues?

“Have the same eigenvalues” shall be interpreted as “have the same **multiset** of eigenvalues.”

Answer: No, they do not have the same eigenvalues.

Although 1,2 appear in both, the **multiplicities** differ:

matrix A has eigenvalue 1 with multiplicity 2

while B has it with multiplicity 1

Space $\mathbb{C}$

Complex number written as $z = a + bi$, where $i = \sqrt{-1}$

Complex conjugate of z [z 的共轭复数] is $\bar{z} = a - bi$

Modulus of z [z 的模] is $|z| = \sqrt{a^2 + b^2}$ [using a, b]
 $= z \bar{z}$ [using $z, \bar{z}$]

A complex number $z = a + bi$, where $i = \sqrt{-1}$

Linear Space over Complex Domain

Recall:

Definition of Linear space over $\mathbb{R}$.

A linear space over $\mathbb{C}$ is defined in a similar way:

just change " $\in \mathbb{R}$ " to " $\in \mathbb{C}$ "

For completeness, we state the full definition in the next slide.

Reading: Def: Linear Space over Complex Domain

Definition 23.3 (Linear space over $\mathbb{C}$)

Suppose V is a set associated with two operations:

- (i) Addition "+": $\mathbf{u} + \mathbf{v} \in V, \forall \mathbf{u} \in V, \mathbf{v} \in V$.
- (ii) Scalar multiplication: $\alpha \mathbf{u} \in V, \forall \alpha \in \mathbb{C}, \mathbf{u} \in V$.

V is called a linear space over $\mathbb{C}$ if the 8 axiom axioms hold:

(A1) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}, \forall \mathbf{u}, \mathbf{v} \in V$.

(A2) $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + \mathbf{v} + \mathbf{w}, \forall \mathbf{u}, \mathbf{v}, \mathbf{w} \in V$.

(A3) There exists a element $\mathbf{0}$ s.t. $\mathbf{u} + \mathbf{0} = \mathbf{u}, \forall \mathbf{u} \in V$.

(A4) If $\mathbf{u} \in V$, then there exists $-\mathbf{u} = (-1)\mathbf{u}$, s.t. $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$.

(A5) $\alpha(\mathbf{u} + \mathbf{v}) = \alpha\mathbf{u} + \alpha\mathbf{v}, \forall \alpha \in \mathbb{C}, \mathbf{u}, \mathbf{v} \in V$.

(A6) $(\alpha + \beta)\mathbf{u} = \alpha\mathbf{u} + \beta\mathbf{u}, \forall \alpha, \beta \in \mathbb{C}, \mathbf{u} \in V$.

(A7) $\alpha(\beta\mathbf{u}) = (\alpha\beta)\mathbf{u}, \forall \alpha, \beta \in \mathbb{C}, \mathbf{u} \in V$.

(A8) $1\mathbf{u} = \mathbf{u}$.

e.g. $\mathbb{C}^n$ is a linear space over $\mathbb{C}$. NOT space over $\mathbb{R}$
 $\mathbb{R}^n$ is NOT a linear space over $\mathbb{C}$.

Space $\mathbb{C}^n$

A few statements about the complex vector space.

$\begin{bmatrix} 1 \\ i \end{bmatrix}$ and $\begin{bmatrix} i \\ -1 \end{bmatrix}$ are linearly *dependent*

A basis of $\mathbb{R}^n$ is automatically a basis of $\mathbb{C}^n$.

e.g. $\{e_1, e_2, e_3\}$ is a basis of $\mathbb{R}^3$ and also a basis of $\mathbb{C}^3$.

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = z_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + z_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + z_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

If $\mathbf{z} = (z_1, \dots, z_n) \in \mathbb{C}^n$, then $\|\mathbf{z}\|^2 = \bar{\mathbf{z}}^T \mathbf{z} = |z_1|^2 + \dots + |z_n|^2 \geq 0$.

Corollary: $\mathbf{z} = \mathbf{0}$ iff $\bar{\mathbf{z}}^T \mathbf{z} = 0$.

$\downarrow$
modulus squared

$$(-1)^2 = 1$$

Example 1: Do Eigenvectors Form a Basis?

Example

$$A = \begin{bmatrix} 3 & 2 \\ 3 & -2 \end{bmatrix}$$

Two eigenvalues $\lambda_1 = 4$ and $\lambda_2 = -3$

(1). For $\lambda_1 = 4$,

For any $t \neq 0$, $x = t \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ is an eigenvector corresp. to $\lambda_1 = 4$

(2). For $\lambda_2 = -3$,

For any $t \neq 0$, $x = t \begin{bmatrix} 1 \\ -3 \end{bmatrix}$ is an eigenvector corresp. to $\lambda_2 = -3$

Do $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -3 \end{bmatrix}$ form a basis of $\mathbb{R}^2$?

Yes.

Example 2: Do Eigenvectors Form a Basis?

Example

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

Two eigenvalues $\lambda_1 = i$ and $\lambda_2 = -i$

(1). For $\lambda_1 = i$,

For any $t \neq 0$, $x = t \begin{bmatrix} 1 \\ i \end{bmatrix}$ is an eigenvector with respect to $\lambda_1 = i$
($t \in \mathbb{C}$)

(2). For $\lambda_2 = -i$,

For any $t \neq 0$, $x = t \begin{bmatrix} 1 \\ -i \end{bmatrix}$ is an eigenvector with respect to $\lambda_2 = -i$
($t \in \mathbb{C}$)

Do $\begin{bmatrix} 1 \\ i \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -i \end{bmatrix}$ form a basis of $\mathbb{C}^2$? *Yes*

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Is it possible that
for any 2×2 matrix,
its eigenvectors form a
basis of $\mathbb{R}^2$ or $\mathbb{C}^2$?

Example 3: Do Eigenvectors Form a Basis?

Example

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

Eigenvalues: $\det(\lambda I - A) = \begin{vmatrix} \lambda - 1 & -1 \\ 0 & \lambda - 1 \end{vmatrix} = (\lambda - 1)^2 = 0 \implies \lambda = 1$

Eigenvectors are Substitute $\lambda = 1$ into $(\lambda I - A)\vec{x} = \vec{0}$:

$$\begin{bmatrix} \lambda - 1 & -1 \\ 0 & \lambda - 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \iff x_2 = 0$$

The null space (eigenspace) is: $\text{Null}(\lambda I - A) = \left\{ t \begin{bmatrix} 1 \\ 0 \end{bmatrix} \mid t \in \mathbb{C} \right\}$

Can you find two eigenvectors that form a basis of $\mathbb{R}^2$?

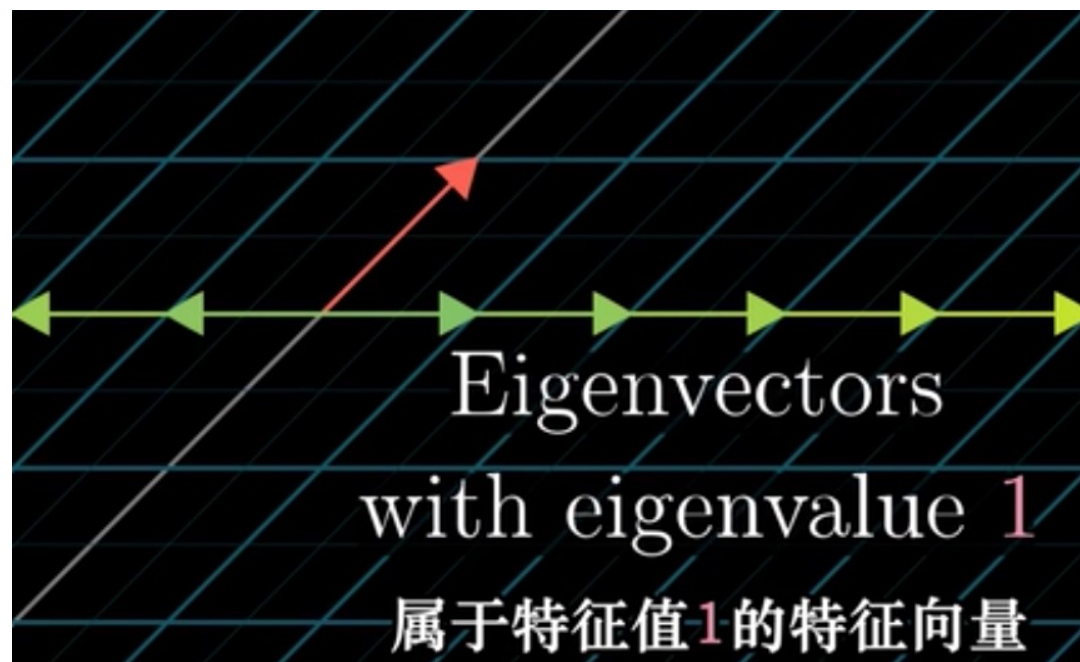
No. All eigenvectors are scalar multiples of $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, so they cannot span $\mathbb{R}^2$ (or $\mathbb{C}^2$).

Remark (advanced reading):

This example is related to **Jordan form**. Interested students may search it; we do NOT teach it.

Eigenbasis may NOT exist

Fact: The (complex) eigenvectors of $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ do NOT form a basis.



Eigenspace dimension $\dim(N(\lambda_1 I - A)) = 1$.
Multiplicity is 2.

Eigenvectors Forming a Basis: Eigenbasis

Definition (Eigenbasis)

A set of eigenvectors of an $n \times n$ matrix that can form a basis of $\mathbb{C}^n$.

Facts:

For some $n \times n$ matrices, the eigenbasis does NOT exist.

(i.e. can find at most $(n-1)$ independent eigenvectors of A)

For some $n \times n$ matrices, the eigenbasis does exist.

(i.e. can find n independent eigenvectors of A)

There is no simple sufficient and necessary condition;

But we will provide a sufficient (but not necessary) condition next.

Part II Spectral Decomposition

Sec. 6.4 of Strang's book v5

45 mins

Conditions for Eigenbasis?

We will show: for an **important class** of matrices, the **eigenbasis exists**;

Not only the eigenbasis exists, but an **orthonormal eigenbasis** exists!

Next, we will:

--first show matrix form of eigen-equations;

Then, we analyze the conditions for the existence of **orthonormal** eigenbasis;

Then, we present the theorem of spectral decomposition.

Part II.1 Matrix Form of Eigen-Equations

8 mins

Preparation: Diagonal Matrix Multiply Matrix

$$D = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}$$

Left multiplication: $A = D \cdot X$

$\Rightarrow$ Multiplies the j -th row of X by λ_j

$$\begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} x_1^T \\ x_2^T \end{bmatrix} = \begin{bmatrix} \lambda_1 x_1^T \\ \lambda_2 x_2^T \end{bmatrix}$$

Right multiplication: $A = X \cdot \underline{D}$

$\Rightarrow$ Multiplies the j -th column of X by λ_j

$$\begin{bmatrix} x_1 & x_2 \\ \downarrow & \downarrow \\ \text{cols} \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = \begin{bmatrix} \lambda_1 x_1 & \lambda_2 x_2 \end{bmatrix}$$

Matrix Form of Eigen-equations

Collection eigen-equations, written in matrix form as follows.

Lemma 23.1 [eigen-equations in matrix form]

Suppose $Av_1 = \lambda_1 v_1, \dots, Av_n = \lambda_n v_n$, where $v_j \in \mathbb{C}^n, \lambda_j \in \mathbb{C}$. Then $AV = VD$, (*)
where $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ is a diagonal matrix, $V = [v_1, \dots, v_n]$.

Proof:

$$\begin{aligned} AV &= A[v_1, \dots, v_n] \\ &= [Av_1, \dots, Av_n] && \text{[whole matrix times column-form]} \\ &= [\lambda_1 v_1, \dots, \lambda_n v_n] && \text{[Def of eigenvalues]} \\ &= [v_1, \dots, v_n] \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} && \text{[property in the previous page:} \\ &= VD. && \text{Multiplying each column by scalars} \\ & && \text{can be written as just ?]} \end{aligned}$$

Lemma 23.2 [reverse: matrix form to eigen-equations]

Consider $D = \text{diag}(\lambda_1, \dots, \lambda_n)$, a diagonal matrix and $V = [v_1, \dots, v_n]$ for $v_j \in \mathbb{C}^n, \lambda_j \in \mathbb{C}$.
If $AV = VD$, then $Av_1 = \lambda_1 v_1, \dots, Av_n = \lambda_n v_n$.

Part II.2 Orthonormal Eigenbasis

15 mins

Orthonormal Eigenbasis

Recall: If a basis of $\mathbb{R}^n$ consists of eigenvectors of a matrix A , we call this basis an “eigenbasis” (corresponding to A).

If the eigenbasis is an **orthonormal basis**, we call this basis an **orthonormal eigenbasis**.

Orthonormal Eigenbasis

Claim 23.1 [orthonormal eigenbasis $\Rightarrow$ symmetric matrix]

Eigenvectors of a real matrix A can form an orthonormal basis of $\mathbb{R}^n$

$\Leftrightarrow A = VDV^T$ where D is a real diagonal, V is a real orthogonal matrix.

Proof:

" $\Rightarrow$ " Suppose eigenvectors $v_1, \dots, v_n$ form an orthonormal basis of $\mathbb{R}^n$.

Then $V = [v_1, \dots, v_n]$ is an orthogonal matrix. ^(invertible) [Def of this type of matrix]

Since v_j 's are eigenvectors, there exists $\lambda_j \in \mathbb{C}$ s.t. $Av_j = \lambda_j v_j, j = 1, \dots, n$.

Since A is a real matrix, $v_j \in \mathbb{R}^n \setminus \{0\}$ (since it's part of a basis), we have $\lambda_j \in \mathbb{R}$.

By Lemma 23.1, we have $AV = VD$, where $D = \text{diag}(\lambda_1, \dots, \lambda_n)$.

$$\text{Then } A = VD \cdot V^{-1} = VDV^T.$$

[re-arranging
previous equation]

[by Property of
orthogonal matrix]

or:

$$(AV)^{V^T} = (VD)V^T$$

"

$$A(VV^T) = A.$$

Orthonormal Eigenbasis

Claim 23.1 [orthonormal eigenbasis $\Rightarrow$ symmetric matrix]

Eigenvectors of a real matrix A can form an orthonormal basis of $\mathbb{R}^n$

$\Leftrightarrow A = VDV^T$ where D is a real diagonal, V is a real orthogonal matrix.

Proof (cont'd):

" $\Leftarrow$ " Suppose $V = [v_1, \dots, v_n]$ is an orthogonal matrix (so its columns form an orthonormal basis of $\mathbb{R}^n$), $D = \text{diag}(\lambda_1, \dots, \lambda_n)$, and $A = VDV^T$. [Def of orthogonal matrix]

We manipulate the equation $A = VDV^T$: multiply both sides by V on the right, we get $AV = VD(V^TV)$
 $= VD$

By Lemma 23.2, we have $Av_j = \lambda_j v_j$ for $j = 1, \dots, n$. $v_j \neq 0, \forall j$ (part of basis)

Thus $\{v_j\}$ are eigenvectors of A that form an orthonormal basis.

Corollary:

If a real matrix A has an orthonormal eigenbasis, then A is a **symmetric matrix**.

$$A^T = (VDV^T)^T = (V^T)^T D^T V^T = V D V^T = A.$$

So far...

$$Av_1 = \lambda_1 v_1, \dots, Av_n = \lambda_n v_n,$$

$$\Leftrightarrow AV = VD, \quad (*)$$

$$Av_1 = \lambda_1 v_1, \dots, Av_n = \lambda_n v_n,$$

and $\{v_1, \dots, v_n\}$ form an orthonormal basis

long,

$$\Leftrightarrow A = VDV^T$$

$\Rightarrow$ A is symmetric.

?

$$A = A^T$$

$\Leftarrow$ What about the other direction? Next: YES!

Part II.3 Spectral Theorem

12 mins

Spectral Theorem

Theorem 23.1 [Spectral Theorem]

Any real symmetric matrix A can be written as

$$A = VDV^T \quad (*)$$

where D is a real diagonal matrix, V is a real orthogonal matrix.

Vector form [practice]:

$$\begin{aligned}
 A &= VDV^T \\
 \begin{matrix} \downarrow \\ n \times n \end{matrix} &= \begin{matrix} \text{column form} \\ \begin{bmatrix} | & | & & | \\ v_1 & v_2 & \dots & v_n \\ | & | & & | \end{bmatrix} \end{matrix} \begin{matrix} n \times n \\ \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} \end{matrix} \begin{matrix} \text{row form} \\ \begin{bmatrix} v_1^T \\ \vdots \\ v_n^T \end{bmatrix} \end{matrix} \\
 &= \begin{matrix} \begin{bmatrix} \lambda_1 v_1 & \dots & \lambda_n v_n \end{bmatrix} \begin{bmatrix} v_1^T \\ \vdots \\ v_n^T \end{bmatrix} \\ \begin{matrix} 1 \times n & n \times n \end{matrix} \end{matrix} \\
 &= \lambda_1 v_1 v_1^T + \dots + \lambda_n v_n v_n^T = \sum_{i=1}^n \lambda_i v_i v_i^T \quad (***)
 \end{aligned}$$

[column & row form]

$\begin{aligned}
 &= \lambda_1 + \lambda_2 + \dots + \lambda_n \quad 1 \times 1 \\
 &= \lambda_1 \|v_1\|^2 + \dots + \lambda_n \|v_n\|^2 \quad 1 \times 1 \\
 &= [\lambda_1, \lambda_2, \dots, \lambda_n] \quad 1 \times n
 \end{aligned}$

(*) and (**): Eigenvalue decomposition (EVD) or eigendecomposition of A .

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exam.

$$[v_1, \dots, v_n] \begin{bmatrix} v_1^T \\ \vdots \\ v_n^T \end{bmatrix}$$

$$= v_1 v_1^T + \dots + v_n v_n^T$$

add

λ_1

$\dots$

λ_n

Spectral Theorem's Corollary

Theorem 23.1 [Spectral Theorem]

Any real symmetric matrix A can be written as

$$A = VDV^T \quad (*)$$

where D is a real diagonal matrix, V is a real orthogonal matrix.

From Claim 23.1, we have:

Coro. 1: Eigenvectors of a real symmetric matrix A can form an orthonormal basis of $\mathbb{R}^n$.

Coro. 2: Eigenvalues of a real symmetric matrix A are real.

Properties of Real Symmetric Matrices

Real Symmetric Matrices

Property 1: All eigenvalues are real.

Property 2: For any eigenvalues, its eigen-space can be spanned by real eigenvectors.

Property 3: Eigenvectors can form an orthonormal basis of $\mathbb{R}^n$.

Together: A can be written as

$$A = V D V^T$$

General Matrices

Eigenvalues can be Complex.
(not real)

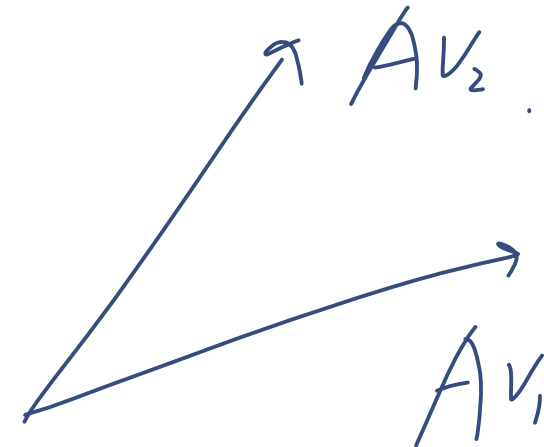
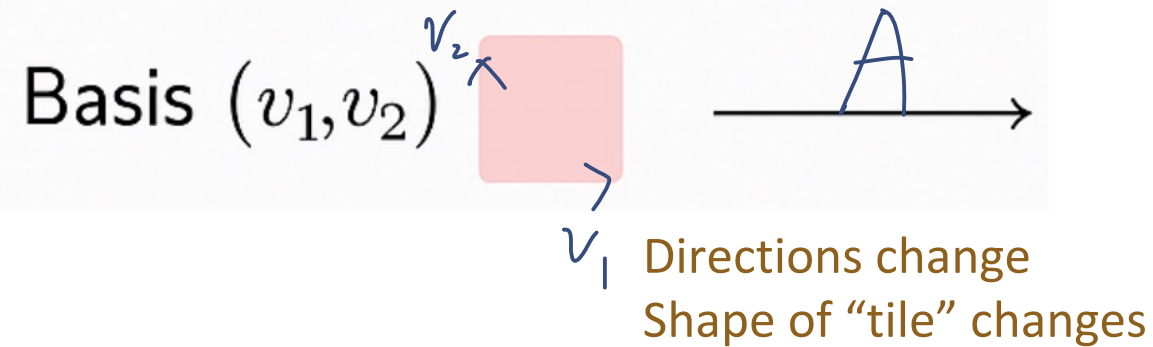
Eigenvectors may all be Complex vectors.
(not real)

Eigenvectors may NOT form basis of $\mathbb{C}^n$.

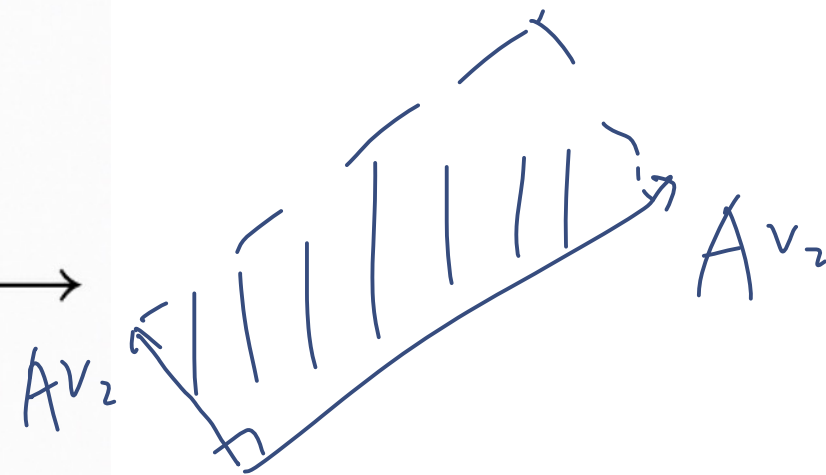
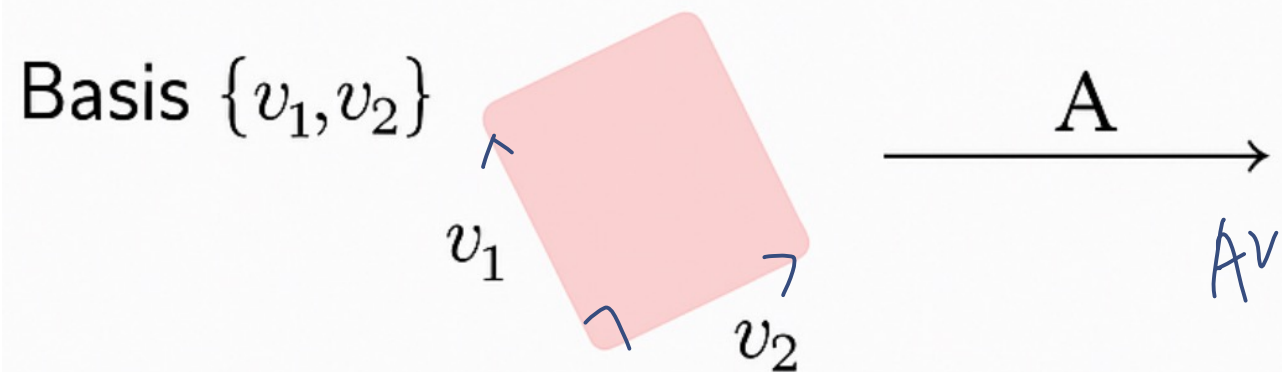
A can be written as?
NOT learned in this class.

Geometry View

Previous lectures



For symmetric $A \in \mathbb{R}^n$



$$x = [v_1, v_2] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$Ax = [v_1, v_2] \begin{bmatrix} \lambda_1 x_1 \\ \lambda_2 x_2 \end{bmatrix}$$

Directions do not change
Shape of "tile": still rectangular

Part II.4 Proof of a Sub- Result

10 mins

Proof of Spectral Theorem

The full proof of spectral theorem is skipped here

Next, we **only** prove one special result
for distinct eigenvalues.

[orthogonal eigenvectors]

Proof of EVD for Distinct Eigenvalues (III): Orthogonality

Assumption 23.1: $A \in \mathbb{R}^{n \times n}$ and $A = A^T$.

Property 23.1 Under Assumption 23.1, eigenvectors corresponding to different eigenvalues are orthogonal.

This property is one part of the spectral theorem.

But our goal is:

use the proof to help you understand a bit of the theorem.

So we can NOT say: by spectral theorem, the property holds.

We want a **proof from scratch**, assuming spectral theorem is NOT known.

Proof of EVD for Distinct Eigenvalues (III): Orthogonality

Assumption 23.1: $A \in \mathbb{R}^{n \times n}$ and $A = A^T$. ①

Property 23.3 Under Assumption 23.1, eigenvectors corresponding to different eigenvalues are orthogonal.

Proof: Suppose $Ax_1 = \lambda_1 x_1$, $Ax_2 = \lambda_2 x_2$, $\lambda_1 \neq \lambda_2$. We need to show $x_1 \perp x_2$.
We compute $x_1^T Ax_2$ in two ways. [computing twice].

$$\begin{aligned}
 & (x_1^T A) x_2 \quad \parallel \quad x_1^T (Ax_2) \\
 & \parallel \textcircled{5} \quad \parallel \textcircled{3} \\
 & (\lambda_1 x_1^T) x_2 \quad \parallel \quad x_1^T (\lambda_2 x_2) \\
 & \parallel \quad \leftarrow \text{scalar anywhere} \\
 & \lambda_1 x_1^T x_2 \quad \parallel \quad \lambda_2 x_1^T x_2
 \end{aligned}$$

$$x_1^T A = ?$$

$$(Ax_1)^T = (\lambda_1 x_1)^T = \lambda_1 x_1^T. \textcircled{5}$$

$$x_1^T A^T \stackrel{\textcircled{1}}{=} x_1^T A$$

$$\Rightarrow \lambda_1 x_1^T x_2 = \lambda_2 x_1^T x_2 \Rightarrow (\lambda_1 - \lambda_2) x_1^T x_2 = 0 \stackrel{\textcircled{4}}{\Rightarrow} x_1^T x_2 = 0, \text{ i.e. } x_1 \perp x_2$$

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$$x_1^T A x_2$$

$$(x_1^T A^T) x_2$$

$$x_1^T (\underline{A x_2})$$

Review Questions

Judgement:

Statement 1: An $n \times n$ real matrix always has n linearly independent eigenvectors.

F. e.g. $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

Statement 2: An $n \times n$ real symmetric matrix always has n linearly independent eigenvectors.

T.

Statement 3: For an $n \times n$ real symmetric matrix, any two eigenvectors are orthogonal.

F.

If v is eigenvector
then $2v$ is also eigenvector
 $v \nparallel 2v$



Summary Today (Write Your Own)

One sentence summary:

Detailed summary:

Summary Today (Instructor)

One sentence summary:

We learned eigenbasis and spectral decomposition.

Detailed summary:

- i) For some matrices, there are n eigenvectors that form a basis, called eigenbasis.

Fact: Eigenbasis may or may NOT exist, for a given matrix.

- ii) Eigenvalue decomposition (特征值分解), or spectral decomposition.

A **real symmetric matrix** A can be written as

$$A = VDV^T = \sum_{j=1}^n \lambda_j v_j v_j^T$$

where

Appendix A Proofs of Spectral Theorem

Proof of EVD for Distinct Eigenvalues (I): Real Eigenvalues

Assumption 23.1: $A \in \mathbb{R}^{n \times n}$ and $A = A^T$.

Property 23.1: Under Assump. 23.1, all **eigenvalues** of A are **real**.

Analysis:

What we know: (i) $A = A^T$ and $A = \bar{A}$.

(ii) Suppose $A\mathbf{v} = \lambda\mathbf{v}$, where $\lambda \in \mathbb{C}$, $\mathbf{v} \in \mathbb{C}^{n \times 1} \setminus \{\mathbf{0}\}$.

What to prove:

Want to prove: $\lambda \in \mathbb{R}$.

Manipulate conditions:

$$A = A^T \text{ and } A = \bar{A} \Rightarrow \boxed{} = \bar{\mathbf{v}}^T \bar{A}^T = \boxed{}$$

Trick: Use two ways to compute $\bar{\mathbf{v}}^T A\mathbf{v}$.

First way: $\bar{\mathbf{v}}^T A\mathbf{v} =$

Second way: $\bar{\mathbf{v}}^T A\mathbf{v} =$

Proof of EVD for Distinct Eigenvalues (II): Real Eigenvectors

Assumption 23.1 (1st part): $A \in \mathbb{R}^{n \times n}$.

Property 23.2: For any eigenvalue of A , there exists a real eigenvector associated to this eigenvalue.

Proof of Property 23.2:

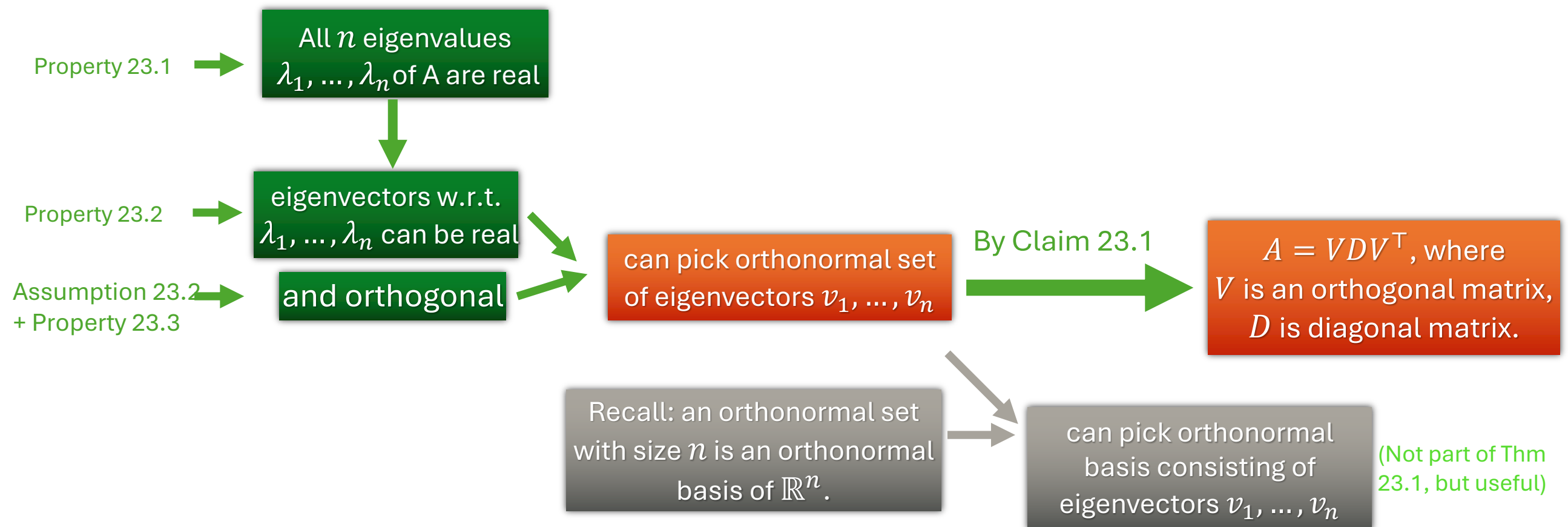
Suppose $A\mathbf{v} = \lambda\mathbf{v}$, where $\lambda \in \mathbb{R}$, $\mathbf{v} \in \mathbb{C}^{n \times 1} \setminus \{\mathbf{0}\}$.

EVD for Distinct Eigenvalues

Assumption 23.1: $A \in \mathbb{R}^{n \times n}$ and $A = A^T$.

Assumption 23.2: all eigenvalues of $A \in \mathbb{R}^{n \times n}$ are distinct.

Proof of Thm 23.1 under Assumption 23.2 (show that $A = VDV^T$ with V orthogonal, D real diagonal)



So far, we **have** proved Thm 23.1 for **distinct eigenvalues**.

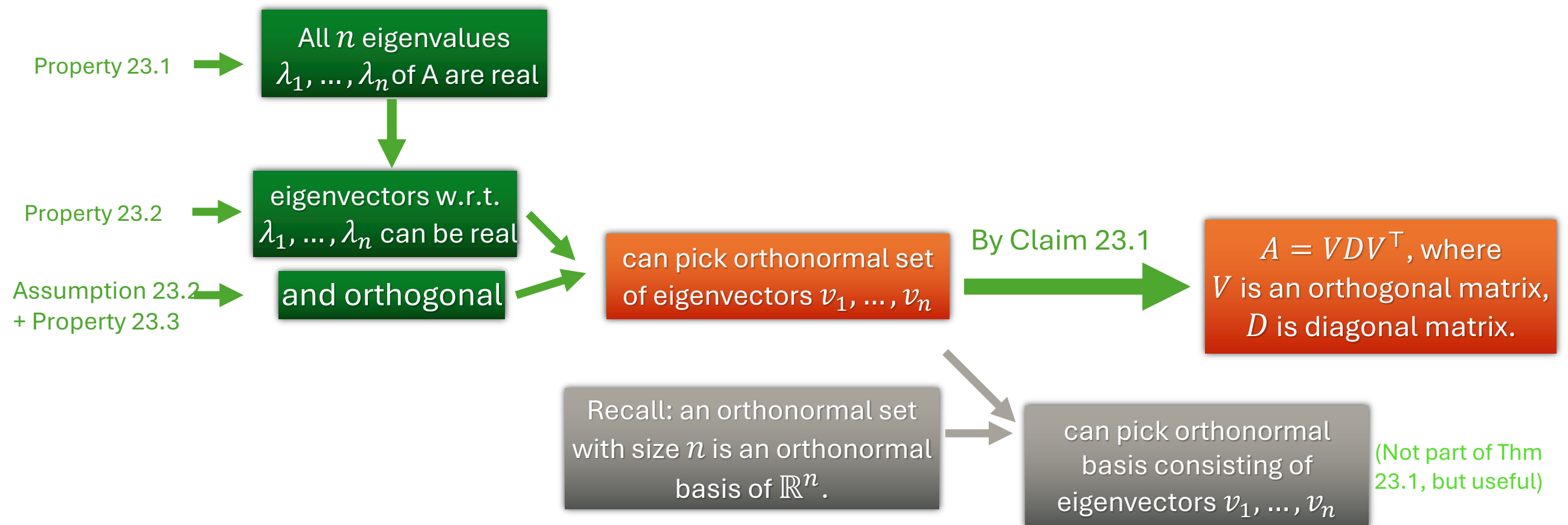
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